

CSCI4390-6390 Assign6

Assign6: Clustering: Density and Spectral

Due Date: Dec 4, before midnight (11:59:59PM)

You will use the [Iris Dataset]((https://archive.ics.uci.edu/dataset/53/iris)). Do not use the last column, which denotes the class label. It will only be use to test the NMI value. The other columns denote the input variables for clustering. For the clustering methods below, it is a good idea to scale all attributes to be within the range 0 to 1. For this you should use [sklearn.preprocessing.MinMaxScaler](#).

CSCI4390/6390: Density Clustering

Implement the DENCLUE density-based clustering algorithm given in Algorithm 15.2 on page 388.

Use mean-shift updates to compute the attractor for each point. Use logsumexp as needed.

Once you have computed the attractors, for step 7 where you check for density reachable, you should basically merge two attractors (and all points attracted to them) using a distance threshold θ . That is, two attractors that are within θ distance of each other should be merged together into a single cluster. Thus, each cluster will be given by a set of attractors (along with all points attracted to any one of them).

Note that this method does not use the number of clusters as input, so you need to choose θ carefully to report the final clustering. Run your code using different thresholds and report the clustering with the best NMI score.

Your program output should consist of the following information:

- Number of clusters.
- Size of each cluster.
- The Normalized Mutual Information (NMI) score for your clustering. You can use [sklearn.metrics.normalized_mutual_info_score](#) using 'geometric' as average_method.

Note that you should try to obtain 3 clusters.

CSCI6390 Only: Spectral Clustering

In addition, implement the spectral algorithm 16.1 on pg 406. To define the similarity matrix, use the Gaussian kernel (see Eq 5.10):

$$a_{ij} = \exp(-(\mathbf{x}_i - \mathbf{x}_j)/2\sigma^2)$$

where σ^2 is the variance parameter, that you have to choose on your own. Run your code using different σ^2 values and report the clustering with the best NMI score. After you obtain the eigenvectors, you can use the [sklearn.cluster.KMeans](#) to run the k-means algorithm to get the final clusters.

Your program output should consist of the following information:

- Number of clusters.
- Size of each cluster.
- The Normalized Mutual Information (NMI) score for your clustering.

What to submit

- Submit your notebook named as **assign6.ipynb** via submitty.

Policy on Academic Honesty

You are free to discuss how to tackle the assignment, but all coding must be your own. Please do not copy or modify code from anyone else, including code on the web. Any students caught violating the academic honesty principle will get an automatic F grade on the course and will be referred to the dean of students for disciplinary action. Use of AI tools must be documented.

Contents © 2023 [Mohammed J. Zaki](#) - Powered by [Nikola](#)